

YongHwi Song

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EDUCATION

Mar. 2023 ~ Present	Chungbuk National University School of Information and Communication Engineering <i>Bachelor Student</i> GPA: 3.83 / 4.5	Cheongju, Korea
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RESEARCH INTERESTS

- World Models and World Action Models (sync test)
- Agentic AI and Tool-Using Language Models
- Large and Small Language Models (LLMs/sLMs)
- Multimodal Learning and Multimodal Foundation Models

RESEARCH EXPERIENCES

- Research Intern at Artificial Intelligence Laboratory (AI Lab), Chungbuk National University, Korea (Jun. 2026 ~ Present) / Conducting research on agentic AI, tool-using language models, and efficient world/world-action models.
- Researcher at Hello, World! Season 2 – World Models Paper Review, Pseudo Lab (가짜연구소), Korea (Oct. 2026 ~ Present) / Reviewing World Models research papers, participating in technical discussions, and contributing structured review documents to the project website.
- Research Intern at Multimedia Information Processing (MIP) Laboratory, Chungbuk National University, Korea (Mar. 2025 ~ Jun. 2026) / Research on Autonomous Driving and Multimodal Reinforcement Learning.
- Advanced Division Leader at Applied Machine Learning Division, HyperCore AI & Data Analytics Club, Korea (Jan. 2026 ~ Present) / Led machine learning study sessions and supported AI project development.

PUBLICATIONS & MANUSCRIPTS

Under Review

- Song, YongHwi*
“K-PragShift: Diagnosing Korean Pragmatic Revisions in Tool-Using Language Models.”
Under review at ACL Rolling Review (ARR), August 2026.

- Song, YongHwi*
“Shapley-style Contribution-Guided Sensor Fusion for Robust Reinforcement Learning-Based Autonomous Driving.”
Submitted to KIISE Transactions on Computing Practices (KTCP), 2026.

In Preparation

- Song, YongHwi*
“Lightweight Action Readout from Multi-Level World Representations for Efficient DriveWAM.”
Ongoing research toward a manuscript, 2026.
 - Song, YongHwi*
“A Comparative Survey of Domestic and International Reinforcement Learning Research for Autonomous Driving.”
Manuscript in preparation, 2026.
- *First author.

CONFERENCE PRESENTATIONS

- Song, YongHwi*
“A Decision Model as a World Model: Comparing JEV with Generative LLMs for LLM Agents.”
Submitted to the 2026 Fall Conference of the Korean Institute of Intelligent Systems (KIIS), 2026.

PROJECTS

- Lightweight Action Readout from Multi-Level World Representations for Efficient DriveWAM, Research Project, Korea / Investigating efficient action prediction by extracting lightweight action representations from multi-level world representations. (Jan. 2026 ~ Present)
- Shapley-guided Multimodal Reinforcement Learning for Autonomous Driving, Faculty-Supervised Research Project, Korea / Implemented Shapley-guided multimodal fusion using RGB, LiDAR, route, and ego-state inputs. (Jun. 2025 ~ Jun. 2026)
- Multi-Agent Vehicle Trajectory Prediction on Argoverse 2, Personal Project, Korea / Implemented trajectory prediction models using the Argoverse 2 Motion Forecasting dataset. (Mar. 2026 ~ Jun. 2026)
- Folding — Local-First Document Agent, Team Project, Korea / Developed a local-first document agent for retrieving, connecting, and safely editing heterogeneous documents. (Jul. 2026)
- A Comparative Survey of Domestic and International Reinforcement Learning Research for Autonomous Driving, Independent Research Project, Korea / Analyzed 93 RL-based autonomous driving studies and summarized key research trends. (Sep. 2025 ~ Present)
- Solar TutorBoard: AI-Powered Personalized Learning Platform, Upstage MixUp Agent Hackathon, Korea / Built AI-agent workflows and backend infrastructure for an educational AI platform. (May 2026)

COMPETITIONS & HACKATHONS

LG Aimers 9th Cohort — KBO Pitch Control Success Probability Prediction AI Hackathon

- Advanced to Phase 2 Online AI Hackathon (Top 20%). Worked on predicting pitch-control success probabilities from game context, player history, and tracking data. (Aug. – Sep. 2026)

2026 National Institute of Korean Language AI-Malpyung Competition — Argumentative Writing Scoring

- Ranked 23rd in the preliminary competition and advanced to the 50-team final arena. Developed an AI system for automated evaluation of Korean argumentative writing. (Jul. – Sep. 2026)

2026 CODEGATE AI Startup Hackathon — Finalist

- Selected as one of approximately 20 finalist teams for the offline hackathon and demo day. Participated in intensive development and validation of an AI-based startup MVP. (Jul. 2026)

Upstage × BDAI Agent Development Hackathon — Finalist

- Solar TutorBoard / Developed a Solar Pro3-based multi-agent platform for tutoring operations, including lesson reports, payment reminders, and schedule coordination. (2026)

AWARDS AND HONORS

- Undergraduate Research Assistant Scholarship, Chungbuk National University, Korea (Sep. 2025 ~ Mar. 2026)
- Academic Excellence Scholarship, Chungbuk National University, Korea (Mar. 2026)
- Academic Excellence Scholarship, Chungbuk National University, Korea (Mar. 2024)
- Academic Excellence Scholarship, Chungbuk National University, Korea (Sep. 2023)

ADDITIONAL PUBLICATIONS

1. Hyeban Namgung, Kuk Seorin, Park Geonwoo, Bae Seohyun, Song Yonghwi, Lee Hyesong, Han Jian, Hyojong Kim, “A Comparative Study on the Prevalence of Forensic Flies Using Chicken Corpse”, Journal of Science Education for the Gifted, (2020)

SKILLS AND TECHNIQUES

- Programming:
Python, C/C++, Java
- Machine Learning Frameworks:
PyTorch, TensorFlow, scikit-learn, Stable-Baselines3
- World Models & Reinforcement Learning:
World Models, World Action Models, Reinforcement Learning, Model-Based Reinforcement Learning, Actor-Critic Methods, Multimodal Policy Learning
- LLMs & Agentic AI:
Large/Small Language Models (LLMs/sLMs), Agentic AI, Tool-Using Language Models, Multi-Agent Systems, Structured LLM Outputs, Local LLM Inference
- Multimodal & Embodied AI:
Multimodal Learning, Sensor Fusion, Computer Vision, Embodied AI, Physical AI, Trajectory Prediction
- Autonomous Driving & Simulation:
CARLA, Argoverse 2, ManiSkill, Motion Forecasting, Autonomous Driving, Robotic Manipulation
- Development & AI Systems:
Git, Linux, FastAPI, Supabase, Ollama, LM Studio, REST APIs